SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks:50

80%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Dharani Nikesh.N.S (192424201) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dharani Nikesh N S

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

<u>Task</u>

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

ANSWER:

1. IPv4 Subnet Misconfiguration Debugging

Configured Parameters:

• Workstation IP Address: 192.168.10.65

• Subnet Mask: 255.255.255.192 (/26)

• Default Gateway: 192.168.10.1

• Target Network Subnet: 192.168.10.0/26

Detailed Technical Analysis

1. Subnet & Range Identification: A /26 mask has a block size of 64 addresses (2^(32-26) = 64). For the IP address 192.168.10.65, the network block spans from 192.168.10.64 to 192.168.10.127. The network address is **192.168.10.64**, the broadcast address is **192.168.10.127**, and the valid host range is **192.168.10.65 – 192.168.10.126**.

2. Error Diagnostics: The configured IP address (192.168.10.65) is a valid host address within its subnet. However, the configured default gateway (192.168.10.1) belongs to the adjacent subnet (192.168.10.0/26, range .1–.62). Because the host and gateway reside in different broadcast domains, local ARP requests for the gateway fail or require inter-VLAN routing, breaking communication with other systems on 192.168.10.0/26.

3. Optimization & Correction: To join the target 192.168.10.0/26 subnet alongside gateway 192.168.10.1, reassign the host an available IP in the 192.168.10.1–192.168.10.62 range (e.g., **192.168.10.10**). Total usable host addresses in the corrected subnet: **62 hosts** (2^6 - 2).

Parameter	Configured (Faulty)	Corrected (Subnet 192.168.10.0/26)
IP Address	192.168.10.65	192.168.10.10 (Valid host)
Subnet Mask	255.255.255.192 (/26)	255.255.255.192 (/26)
Network Address	192.168.10.64	192.168.10.0
Broadcast Address	192.168.10.127	192.168.10.63
Valid Host Range	192.168.10.65 – 192.168.10.126	192.168.10.1 – 192.168.10.62
Default Gateway	192.168.10.1 (Out-of-subnet)	192.168.10.1 (In-subnet)
Usable Host Capacity	62 addresses	62 addresses

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

ANSWER:

2. IPv6 Address Expansion & Connectivity Debugging

Configured IPv6 Addresses:

• Device A: 2001:db8::ff00:42:8329/64

• Device B: 2001:db8:0:0:1::1/64

Expanded Notation & Subnet Comparison

1. **Address Expansion (128-bit Full Representation):**
   - **Device A:** 2001:0db8:0000:0000:0000:ff00:0042:8329/64
   - **Device B:** 2001:0db8:0000:0000:0001:0000:0000:0001/64

2. Prefix & Subnet Evaluation:

- **Device A Prefix (First 64 bits):** 2001:0db8:0000:0000::/64 (2001:db8:0:0::/64)
- **Device B Prefix (First 64 bits):** 2001:0db8:0000:0000::/64 (2001:db8:0:0::/64)
- **Same Subnet Verification:** Yes, both devices share the exact same 64-bit network prefix. There is no IP prefix mismatch.

3. Root Cause Analysis for Failure:

Since IP addressing is mathematically valid and both hosts reside on 2001:db8::/64, communication failure is caused by lower-layer or protocol-level issues:

- **Neighbor Discovery Protocol (NDP) Blocked:** ICMPv6 Type 135 (NS) and Type 136 (NA) packets are blocked by host firewalls or ACLs.
- **VLAN Isolation:** Switch ports are assigned to different access VLANs without L2 adjacency.
- **Complex Interface Notation:** Device B contains '1' in the 5th hextet (part of the 64-bit Interface ID, not the network prefix), which can cause visual confusion during manual setup.

4. Optimization & Capacity:

Simplify addressing using clean sequential interface IDs (e.g., Device A = 2001:db8::1/64, Device B = 2001:db8::2/64). Available host addresses in this /64 subnet: 2^64 = 18,446,744,073,709,551,616 addresses (IPv6 does not use broadcast addresses).

Device	Expanded IPv6 Address (128-bit)	64-Bit Network Prefix	Recommended Clean IP
Device A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:db8:0:0::/64	2001:db8::1/64
Device B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:db8:0:0::/64	2001:db8::2/64

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

ANSWER:

3. Variable Length Subnet Masking (VLSM) Optimization

Initial Base Network: 192.168.1.0/24 (Total 256 addresses)

Existing Inefficient Allocation Analysis:

Subnet Prefix	Network Address	Broadcast Address	Valid Host Range	Usable Hosts	Efficiency Assessment
/26	192.168.1.0	192.168.1.63	192.168.1.1 – .62	62	High waste if allocated to small team
/27	192.168.1.64	192.168.1.95	192.168.1.65 – .94	30	Adequate for medium team
/28	192.168.1.96	192.168.1.111	192.168.1.97 – .110	14	Risk of address exhaustion

Root Cause & Debugging Analysis

• **Flawed Static Allocation:** Static mask assignment creates rigid subnet boundaries without considering actual host density. Assigning a /28 (14 usable IPs) to a growing team causes early IP depletion, while assigning a /26 (62 usable IPs) to a small department leaves dozens of IPs permanently unused.

• **Discontiguous Allocation Gaps:** Arbitrary subnetting without top-down planning leaves unallocated address holes that cannot be aggregated easily.

Recommended VLSM Optimization Plan

To optimize utilization, allocate subnets strictly in descending order of size based on actual host requirements:

Department Needs	Allocated CIDR	Subnet Mask	Network Address	Host Range	Broadcast	Usable Hosts
Dept A (Large: 50	/26	255.255.255.192	192.168.1.0	.1 – .62	192.168.1.63	62

hosts)						
Dept B (Medium: 25 hosts)	/27	255.255.255.224	192.168.1.64	.65 – .94	192.168.1.95	30
Dept C (Small: 10 hosts)	/28	255.255.255.240	192.168.1.96	.97 – .110	192.168.1.111	14
Future Expansion / Unused	/25	255.255.255.128	192.168.1.128	.129 – .254	192.168.1.255	126

4. Two devices are configured with the following IPv6 addresses:

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Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

ANSWER:

4. IPv6 Address Expansion & Connectivity Debugging

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Device B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:db8:0:0::/64	2001:db8::2/64

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

<u>Task</u>

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

ANSWER:

5. ISP Routing Table & Longest-Prefix Match Analysis

Routing Table State:

• 192.168.1.0/24 -> Next Hop: 10.0.0.2

• 192.168.2.0/24 -> Next Hop: 10.0.0.3
• Default Route (0.0.0.0/0) -> Next Hop: 10.0.0.1

Target Packet Destination: 192.168.2.100

Routing Process & Debugging Steps

1. Longest Prefix Match (LPM): The router compares destination IP 192.168.2.100 against all routing table entries. Both 0.0.0.0/0 (/0) and 192.168.2.0/24 (/24) match the packet, but the router selects **192.168.2.0/24** because /24 is the longest, most specific bit-prefix match.

2. Next-Hop Selection: The router forwards packets targeting 192.168.2.100 to next-hop gateway **10.0.0.3**.

3. Debugging Packet Delivery Failure: Since routing lookup succeeds theoretically, reported packet loss to 192.168.2.100 indicates one of the following underlying issues:

• **Unreachable Next Hop:** Gateway 10.0.0.3 is offline, ARP resolution for 10.0.0.3 fails, or the local interface is down.

• **Missing Return Path (Asymmetric Routing):** The destination host or gateway 10.0.0.3 lacks a return route back to the source subnet.

• **Access Control Filtering:** An outbound ACL on the local router or an inbound firewall rule on 10.0.0.3 drops the traffic.

4. Fallback Handling: If a packet is sent to an unknown IP (e.g., 8.8.8.8), no specific /24 route matches. The router falls back to the Default Route (**0.0.0.0/0**) and forwards packet to gateway **10.0.0.1**.

Lookup Stage	Router Decision / Result	Debugging Assessment
Target Destination IP	192.168.2.100	Valid unicast IP in 192.168.2.0/24
Matching Route	192.168.2.0/24	Selected via Longest-Prefix Match (/24 vs /0)
Forwarding Gateway	10.0.0.3	Next-hop interface must be active
Failure Diagnostic	L3 Routing Lookup Succeeded	Check L2 ARP, gateway 10.0.0.3 health, or return routes
Unmatched IP Fallback	0.0.0.0/0 -> 10.0.0.1	Standard default gateway path

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly

		understanding(2)			
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification(1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation

	Problem Identification (20%)	Debugging (20%)	Optimization (25%)	Analysis (20%)	Code Quality (15%)	
Q1	1.5	1.5	2	1.5	1	7.5
Q2	1.5	1.5	2	1.5	1	7.5
Q3	2	2	1.5	2	1.5	9
Q4	1	2	2.5	2	1	8.5
Q5	1	1.5	2	1.5	1.5	7.5
						40